Human hairs to control oil spills

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ABSTRACT:

Oil pollution in wastewater can occur in a variety of ways, including oil spills, the discharge of untreated oil and gas production waste, and runoff from oil-contaminated regions. When oil is spilled into water, it may cause severe environmental damage as well as harm to aquatic ecosystems and human health. Here are a few examples of how oil pollution can harm water quality and the environment:

1. Toxicity: Oil spills can contaminate water with hazardous substances such as benzene and other carcinogens. These compounds have the potential to damage aquatic plants and animals, as well as humans who come into contact with polluted water.
2. Oxygen depletion: Oil spills can decrease the amount of oxygen in the ocean. This has the potential to kill fish and other aquatic species that rely on oxygen to thrive.
3. Physical smothering: Large oil spills can literally suffocate and kill aquatic species by covering their gills, feathers, or hair, preventing them from breathing or thermoregulating.
4. Bioaccumulation: Over time, oil may build up in the tissues of fish and other aquatic species, leading to bioaccumulation in the food chain. This can endanger the health of both animals and humans that ingest tainted seafood.
5. Degraded environments: Oil pollution may harm critical habitats such as coral reefs, wetlands, and estuaries, which are essential for the survival of many aquatic species.
6. Many efforts may be taken to avoid and clean up oil pollution in water, including tightening industry rules and standards, employing effective spill prevention and response tactics, and promoting alternative energy sources. These measures can assist to limit the amount of oil discharged into the environment and the adverse consequences of oil pollution on human health and the ecosystem.
7. The Exxon Valdez oil spill occurred in the Prince William Sound, Alaska, on March 24, 1989. Exxon Valdez, an oil supertanker owned by Exxon Shipping Company bound for Long Beach, California struck Prince William Sound's Bligh Reef, 1.5 mi (2.4 km) west of Tatitlek, Alaska at 12:04 a.m. and spilled 10.8 million US gallons (260,000 bbl) (or 37,000 tonnes) of crude oil over the next few days

Introduction:

Oil spills occur during petroleum exploration, production, and transportation, causing ecological damage. They jeopardize the environment, human and animal health, and natural resources. Cleanup methods include burning, dispersants, skimmers, booms, and absorbents. Synthetic sorbents are expensive and non-biodegradable, while natural alternatives like kapok, rice husk, and coconut husk offer effectiveness, affordability, eco-friendliness, and high sorption capacity. Human hair, with its water-repellent cuticle and porous cortex, is a natural absorbent made up of proteins, water, lipids, and trace elements. It contains peptide bonds and groups that form hydrogen bonds between neighboring hair molecules. During the Exxon Valdez disaster, a hairstylist named McCrory proposed using human hair for oil spill cleanup. However, there is limited research on this topic. This study aims to investigate how well human hair can absorb oil spills and how different hair types and structures affect its effectiveness.

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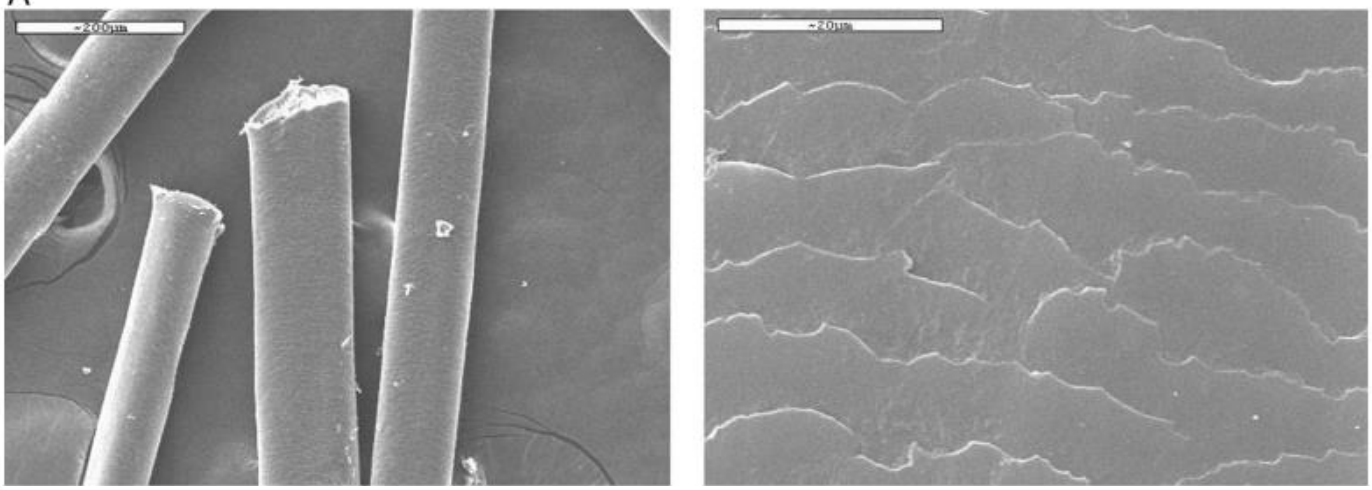

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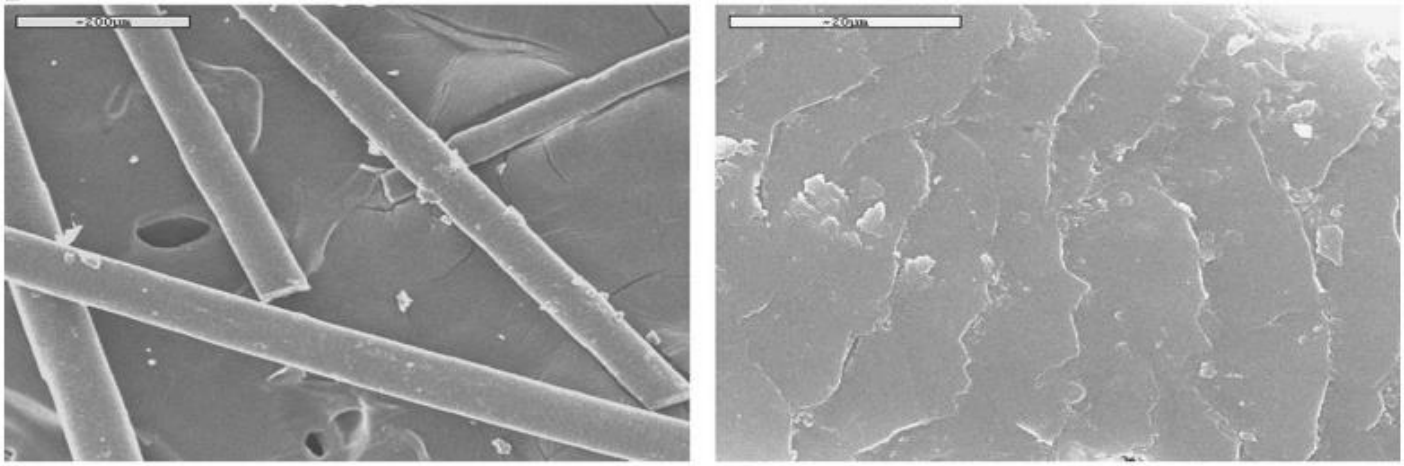

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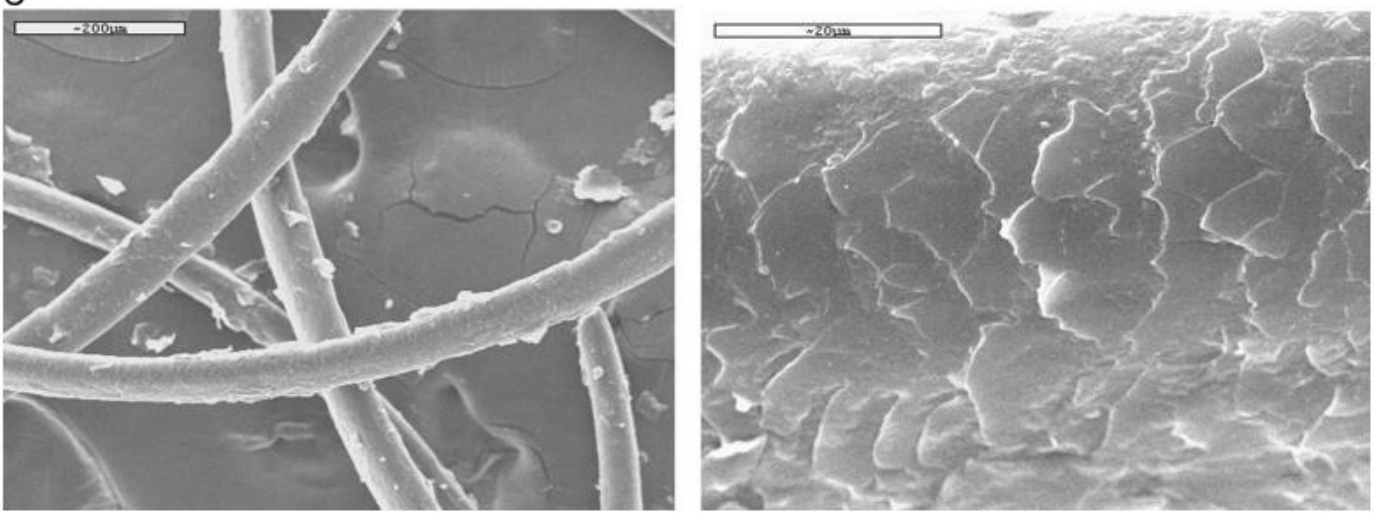


WHY USE HAIR?

- Hair is an effective natural absorbent that has been used for many years to clean up oil spills. It is a renewable and sustainable resource that is readily available from various sources, such as salons, pet groomers, and individual donations. When woven or stuffed into mats, booms, or pillows, hair can absorb up to 50 times its weight in oil, making it ideal for oil spill cleanups. The main advantage of this system is that there is no external power source required.

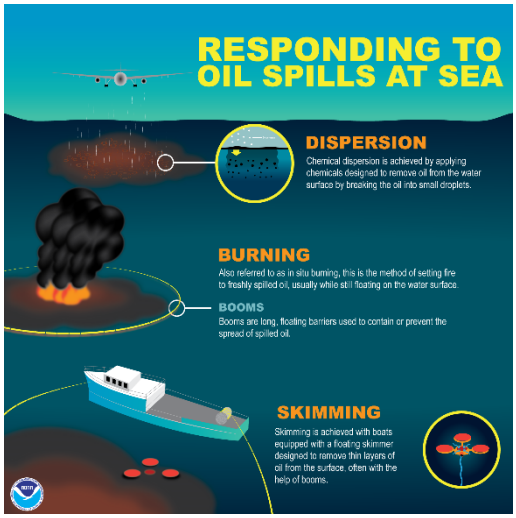
- One of the advantages of using hair for oil spill cleanups is its high affinity for oil. Hair is naturally oily due to the presence of sebum, a waxy substance produced by the sebaceous glands in the skin. This oily nature of hair allows it to attract and absorb oil from spills while repelling water. When hair is used to clean up oil spills, it adheres to the oil, and the oil is trapped within the material, preventing it from spreading further. The future scope of our project is extremely good. It can be applied in any place where ever there is mechanical stress.
- Furthermore, hair is a natural and sustainable resource that does not harm the environment. Unlike synthetic materials, which are non-biodegradable and can cause harm to wildlife and ecosystems, hair-based materials are biodegradable and can be composted or recycled after use. Additionally, hair-based materials are cost-effective and easy to use, making them an attractive option for local communities and organizations involved in environmental cleanups.

Similarly, in 2010, a hairdresser from Alabama, Lisa Gautier, collected hair from salons and stuffed it into nylon stockings to create oil-absorbing booms. These booms were used to clean up the BP oil spill in the Gulf of Mexico, and they were found to be effective in absorbing oil from the water.

Primitive Methods:

1. Skimming: One of the simplest and oldest methods for cleaning up oil spills is skimming. This involves using booms and nets to contain the oil and then skimming the oil off the surface of the water with specialized tools.

   The oil can then be collected and removed.



2. Absorbent materials: Another primitive method for cleaning up oil spills is to use absorbent materials, such as hay, straw, or sawdust. These materials are spread on the surface of the water and then soaked up with the oil.

   The soaked-up material can then be removed and disposed of.

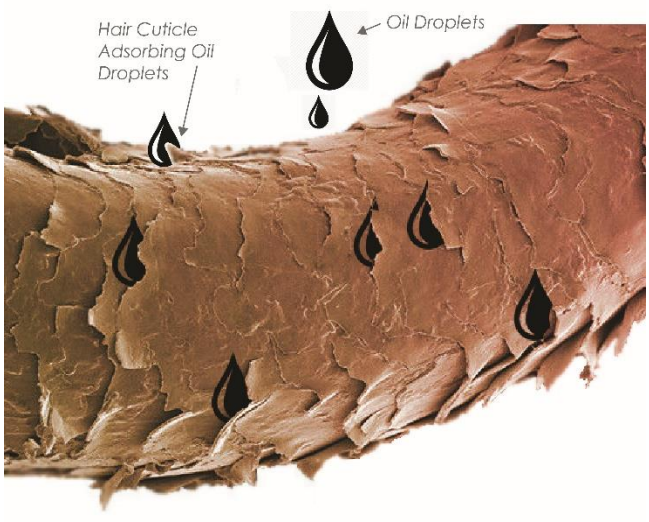
   Polypropylene.

3. Natural degradation: Natural degradation is another primitive method for cleaning up oil spills. This involves allowing the oil to break down naturally over time. This method can be effective for small spills and in areas with a lot of sunlight and oxygen.

4. Burning: Burning the oil on the surface of the water is another primitive method for cleaning up oil spills. This method involves igniting the oil on the surface of the water, which then burns off the oil. However, this method can be dangerous and can create air pollution.

5. It is important to note that these primitive methods may not be as effective as modern methods and may not be appropriate for all types of oil spills. Additionally, some of these methods may have negative impacts on the environment or public health. It is recommended to use modern methods for cleaning up oil spills whenever possible.

How does it work?



- **Hydrophobicity**: Hair is inherently hydrophobic, which means that it repels water. This characteristic also makes hair repellent to oil, which is similarly hydrophobic.
- **Porosity**: Hair is porous, which means it has tiny pores or crevices in its structure. These pores enable the hair to absorb and hold oil, preventing it from spreading.
- **Adsorption**: Hair has the potential to absorb and retain oil molecules via a process known as adsorption. Hair is good in removing oil from the environment due to its feature.
- **Surface** Area: Hair has a high surface area to volume ratio, which indicates it has a huge surface area in comparison to its size. Hair absorbs and traps more oil than other materials of comparable volume due to this characteristic.
- **Availability**: Hair is generally accessible and can be obtained from hair salons, pet grooming facilities, or other sources. As a result, it is a cost-effective and long-term solution for preventing oil leaks.

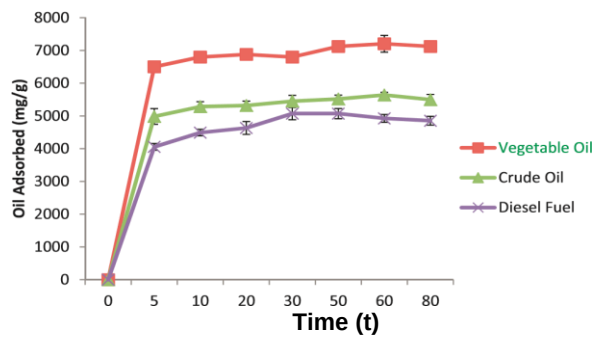
How are hair mats made:

1. Gather hair: Gather hair from hair salons, pet groomers, and other sources. Make sure your hair is clear of hair products and debris.
2. Sort and clean the hair: Separate any longer strands and sort the hair into lengths. Rinse the hair with water to eliminate any dirt, oil, or debris.
3. Set up the mat frame: To keep the hair in place, use a plastic or wire mesh frame. The frame should be somewhat bigger than the mat's intended size.
4. Spread the hair evenly across the frame, overlapping the strands to form a thick mat. Check for any holes between the strands.
5. Sew the hair in place: Using a needle and thread, sew the hair onto the frame. Begin at one end of the frame and work your way across, ensuring that the hair is securely fastened to the frame.
6. Finish the mat: After you've sewed all of the hair onto the frame, remove any loose hairs or excess strands.
7. Dry the mat: Place the mat in a well-ventilated room to dry.
8. Once dried, use the hair mat as an absorbent material to clean up oil spills. When the hair mat has absorbed as much oil as possible, contact a hazardous waste disposal centre or an organisation that specialises in recycling hair mats to properly dispose of it.

PROTOTYPE:

The oils employed in the sorption test were vegetable oil, crude oil and diesel fuel. The characteristics of these oils are outlined in Table. Human hair used for the adsorption experiments were collected from a local salon in Coventry and was segregated according to different types as Sorbent A , Sorbent B and Sorbent C. The particle sizes of the different hair ranged from 60 to 70mm.

Oil type	Density(g/cm)	Viscosity(Pas)
Vegetableoil	0.910	0.061
Motoroil	0.849	0.024
Dieselfuel	0.840	0.005



Cleaning of hair.

The segregated hairs were soaked in hot water and washed several times with detergent. Then, they were rinsed with. hot water and dried in natural sunlight for 48 h. The morphology and cross sections of a sample of each of the three different hair types (A, B and C) were taken a look at by our NITT facilities. An electron microscopic scan was take. using OJAS facilities.

Sorption experiments

Adsorption experiments were conducted in water. 20 g of crude oil, vegetable oil
and diesel fuel, were poured into separate 500 water. Varying amounts of the sorbents (0–8 g) were added wit 60min.
Oil adsorption capacities(mg/g) were obtained from the following equation:

$$\text{Sorption capacity} = \frac{X_O - X_S}{X_S}$$

(1) where X_0 (g) is the total mass of wet sorbent after oil adsorption
(2) XS (g) is the mass of the sorbent before adsorption. All experiments were conducted at room temperature (25 degree C) and were performed in triplicate with the average value and standard deviation (SD) calculated. Sample data with SD greater than 10% were rejected with a re-run of the test carried out. A dosage of 4 g was used for all subsequent experiments.

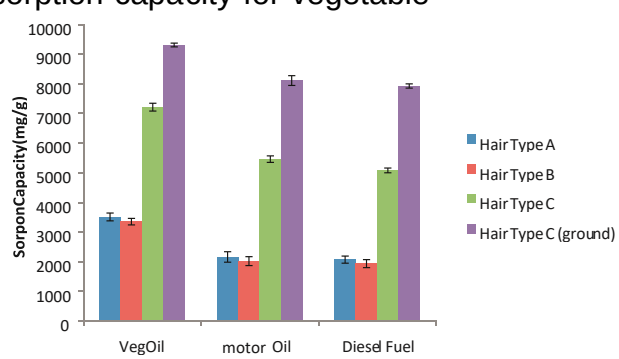
Results and discussion.

Effects of contact time on adsorption

The effects of contact time on the adsorption of human hair were determined by batch adsorption experiments at varying contact times of 5, 10, 20, 30 40, 60 and 80 min with all other experimental variables kept constant. The effects of contact time are illustrated in There was a rapid increase in the sorption of the various oils in the first 5 min and after that, proceeded at a slower rate and finally attained equilibrium before the 60th minute. The initial high rate of oil sorption may be attributed to the existence of bare active sites on the surfaces of human hair. As portions of these sites get occupied with the oil molecules with increasing contact time, the adsorption rate became slower until the 60th minute when there was no more significant adsorption of oil, probably due to the saturation of the surfaces of the hair with oil molecules as well as the equilibrium between the adsorption and desorption processes that occurs after saturation. Similar results were obtained by other researchers regarding the use of biosorbents for oil removal. More than 90% of the total adsorbed oil took place in the first 5 min, indicating that a short contact time is required in possible field application of human hair for oil spill cleanup.

Effects of hair type, particle size and adsorbent dose

Adsorption capacities for the three categories of human hair were determined in vegetable oil, diesel fuel and crude oil by batch adsorption experiments. The adsorption capacities of the three hair types are shown in fig. All the hair types showed the highest sorption capacity for vegetable oil followed by crude oil, with diesel fuel exhibiting the lowest capacity. Type A had a sorption capacity of 3500, 2150 and 2060 mg/g for vegetable oil, crude oil, and diesel fuel, respectively. The respective values for Type B were 3340, 2010 and 1925 mg/g and for the Type C were 7203, 5450 and 5070 mg/g. The superior performance of Type C hair can be attributed to its tight, coarse, and spongy nature. The microstructure and morphology of the different hair types were examined under a scanning electron microscope. shows the microscopic structures at 20 and 200mm. Hair Type C had rougher cuticles formed by many hair layers which look like a scalp. It is more hydrophobic as water tends to bead on the hair strands. Also, there are more macropores, which play a significant role in the sorption of oil by human hair due to capillary action of the oil at the hair/oil interface.Type A and Type B hairs are relatively similar and less tightly packed compared to Type C hair. These characteristics account for the better sorption capacities of Type C hair, which achieved twice the adsorption capacities of Type A and Type B. It was also observed that human hair has more affinity for vegetable oil compared to crude oil and diesel fuel. This is consistent with the findings of Peng et al. [36]. The high adsorption capacity for vegetable oil may be due to its higher viscosity (Table 1) which enhanced the adherence of vegetable oil to the surface of the sorbents compared to crude oil and diesel fuel with lower viscosities [34,37,40]. The effect of particle size modification was investigated by pulverizing Type C hair. The ground hair (particle size 15–25mm) was used for the batch adsorption tests. The adsorption capacity of Type C hair significantly improved to 9300, 8100 and 7917 mg/g for vegetable oil, motor oil and diesel fuel, respectively.



Comparatively, human hair showed a better sorption capacity than natural wool fiber (5560 mg/g for oily wastewater). modified oil palm leaves (646 mg/g for vegetable oil adsorption) and rice husk (6220 mg/g for crude oil). It also compares favorably with the adsorption capacities of nonwoven polypropylene, an expensive commercial sorbent which has been reported to be between 9100 and 10,000 mg/g, making human hair a potential alternative low-cost natural adsorbent to nonwoven polypropylene sorbents. This high sorption capacity of human hair can be attributed to the presence of keratin and cuticles covered by a layer of lipid (18-methyleicosanoic) which is a key to its oleophilic and hydrophobic characteristics, hence the ability to sorb large quantities of oil from aqueous solution. In addition, its porous structure, rough texture, large surface area and the presence of amine (NH), hydroxyl (OH), sulphide group, SS, carboxyl group, CO and carbonyl (C¼O) groups on its surface may have contributed to the high sorption affinity of human hair. The effect of adsorbent dose on the adsorption efficiency of the Type C hair was investigated. Shows the adsorption efficiencies for all three oils with increasing mass of the adsorbent. The percentage uptake for diesel fuel by human hair was found to be lower than vegetable oil and crude oil. This may be linked to the higher viscosities of vegetables and crude oil.

Adsorption isotherms and kinetics

An adsorption isotherm helps to provide an understanding of the interactions between the solute and the adsorbent and also helps in modelling design parameters. The equilibrium data obtained in this study was analyzed using the Langmuir and Freundlich adsorption isotherm models [30,31]. The linear form of the Langmuir isotherm is expressed in Eq. (2) as:

$$\frac{1}{X/M} = \frac{1}{b} + \frac{1}{\mathrm{ab}} \times \frac{1}{C_e}$$

where X – the mass of adsorbate (mg), M – the mass of adsorbent (mg), Ce – the concentration of solute remaining at equilibrium in mg/L, a, b – the constants. a is the coefficient, and b is the amount of adsorbate needed to form a complete monolayer on the adsorbent surface and so increases with molecular size. A plot of (1/(X/M)) against (1/Ce) gave a straight line. Also, the linearized form of the Freundlich isotherm is expressed in Eq. (3)

$$\log\frac{X}{M} = \log k_f + \frac{1}{n}\log C_e$$

where X – the mass of adsorbate (mg), M – the mass of adsorbent (mg), Ce – the concentration of solute remaining at equilibrium in mg/L, kf and n – constants derived from the adsorption isotherm by plotting (X/M) against Ce on log–log paper which produces a straight line with a slope 1/n while the y-intercept is kf. The Langmuir and Freundlich coefficients for single solute adsorption isotherms and their corresponding correlation coefficients are presented in Table 3. The results showed a good fit of the Freundlich model to the experimental data compared to the Langmuir model. This observation is not rare as similar findings have been previously reported [41,43,44]. This indicates that sorption of the oils onto human hair occurred on heterogeneous sites with non-uniform distribution of energy level rather than on a homogeneous site with uniform distribution of energy. The value of constant n, which indicates the intensity of adsorption, showed a favorable sorption of oil onto human hair. Adsorption kinetics helps to predict the rate at which the adsorption

takes place and aids in selecting the optimum operating. conditions in designing and optimising full-scale applications. In this study, the adsorption kinetics of Type C hair for diesel fuel and vegetable oil were investigated by using the pseudo-first order and pseudo-second order adsorption models. Pseudo first order and pseudo second order were proposed by Lagergren and can be expressed, respectively in linear form as:

$$\mathrm{Log}(C_e - C_t) = \mathrm{Log}\, C_e - \frac{K_1 t}{2.303}$$

$$\frac{t}{C_t} = \frac{1}{K_2 C_e^2} + \frac{t}{C_e}$$

where Ce and Ct are the liquid-phase concentrations of oil in mg/l at equilibrium and time t, respectively. k1 and k2 are the first order and second order rate constants, respectively. The pseudo first order plot of Log(Ce Ct) against t should give a linear relationship from which k1 in (min1) can be calculated from the slope of the graph. A plot of t/Ct will give a rate constant k2 (mg g1min1) for pseudo-second order kinetics. presents the constant

Table 3
Constant and co-efficient of determination for Langmuir model and Freundlich equation for Type C hair.

	Langmuir			Freundlich		
	a	b	R^2	n	k_f (g/g) (L/g)$^{1/n}$	R^2
Vegetable oil	0.1320	15.39	0.5834	1.8643	1.0730	0.9648
Diesel fuel	0.0107	20.08	0.8849	2.3798	1.0083	0.9772
Crude oil	1.2556	4.225	0.3428	2.3800	1.4771	0.6247

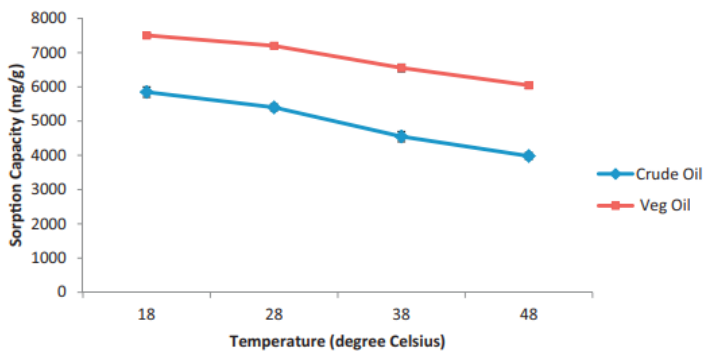
Table 4
Pseudo first and second order rate constants for Type C hair.

Adsorbate	Pseudo-first order kinetic		Pseudo-second order kinetic	
	k_1 (min^{-1})	R^2	k_2 (mg g^{-1} min^{-1})	R^2
Vegetable oil	0.033	0.211	0.0203	0.994
Diesel fuel	0.0203	0.218	0.0387	0.999

valuesand correlation coefficients R2 of both pseudo-first order and pseudo-second order kinetic models for adsorption of diesel fuel and vegetable oil by human hair. The kinetic plots showed a good fit of sorption equilibrium data with respect to pseudo-second order. This suggests that the rate limiting steps may be chemisorption involving valence forces [45].

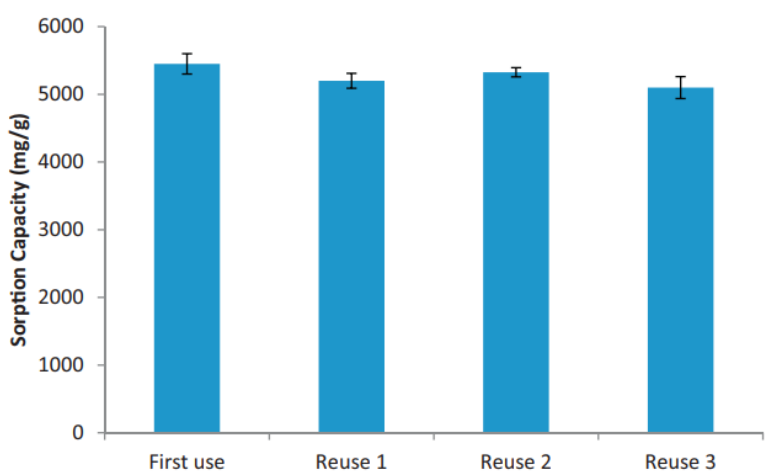
Effects of temperature on adsorption capacity

The effect of temperature on the adsorption capacity of Type C hair in crude oil and vegetable oil were investigated at varying temperatures of 28, 38 and 48. The results are presented in. The results demonstrate that the adsorption capacity of human hair decreases with increasing temperature for both the adsorption of crude oil and vegetable oil. A similar result depicting a decrease in the amount of oil sorbed with temperature increase was reported by Rajakovic et al. while investigating the effect of temperature on the sorption capacity of natural wool fibres and recycled non-woven material. Vlaev et al. using rice husk ash and Sidik et al. using lauric acid modified palm leaves also reported similar behaviour for the adsorption of crude oil. The decrease in adsorption of oil from aqueous media as the temperature of the media increases may be due to the fact that its solubility in water increases, its viscosity decreases, making it drain off the sorbent's surface easily. In addition, its Brownian motion increases, hence increasing the energy required to bind it to the surface of the adsorbent .



Reusability of human hair adsorbent

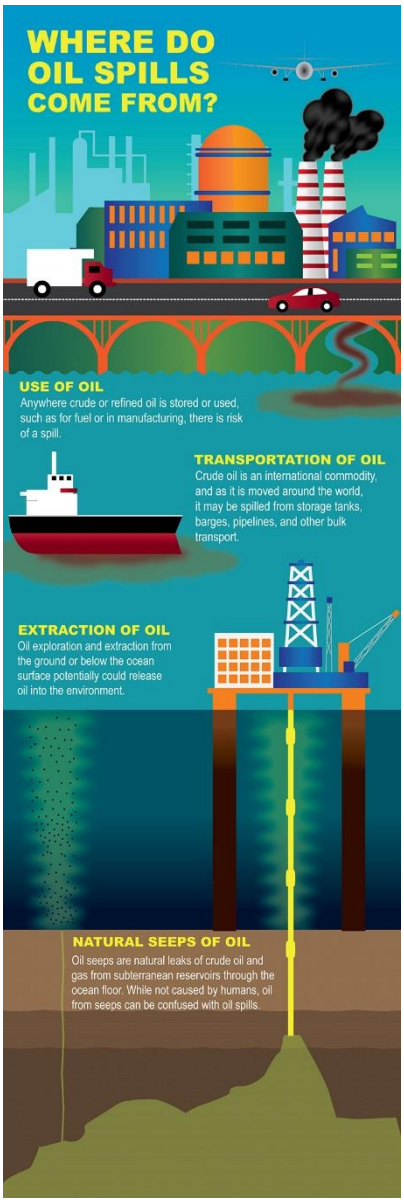
There are different methods in which adsorbents can be recovered and reused, e.g. compression, centrifugation, solvent extraction, but for the purpose of this experiment solvent extraction was employed to evaluate the potential of reusing the human hair without a significant decrease in performance over time. Fig. 6 shows the results of the reusability test for Type C hair in crude oil. There was no significant decrease in the adsorption capacity after four cycles of reuse. This is like the work of Liu et al.using kapok which showed no significant loss in its adsorption capacity after it was subjected to the process of recovery and reuse five consecutive times as well as Lim and Huang four times.



Possible uses:

OIL SPILLS:

Human hair possesses various qualities that make it useful in oil spill recovery efforts.:

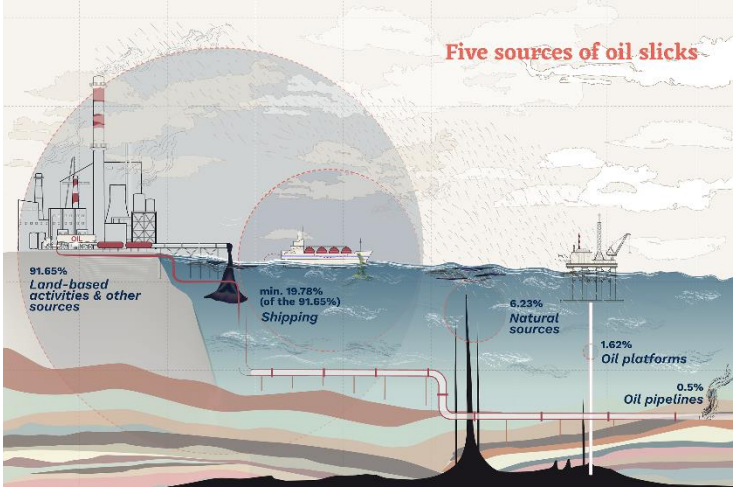


1. **Hydrophobic nature**: Hair has a hydrophobic tendency, which means it repels water while attracting oil. Hair absorbs oil and repels water when used to clean up oil spills, making it an effective absorbent material.
2. **High surface area**: Human hair has a high surface area due to its fine texture and the presence of scales on the hair cuticle. Because of its huge surface area, hair may absorb a big amount of oil per unit weight.
3. **Natural and biodegradable**: Hair is a natural and biodegradable material, which means that it may be easily dissolved by natural processes without harming the environment. As a result, hair is a sustainable and environmentally beneficial choice for oil spill cleanup.
4. **Readily available**: Human hair is widely available in huge numbers, making it an inexpensive and accessible material for use in oil spill cleanup efforts.
5. **Easy to handle**: Hair is simple to handle and carry because it can be compressed into mats or booms for easy storage and deployment.

Overall, human hair is an effective and environmentally safe technique for cleaning up oil spills.

In Industrial Field:

1. Human hair can be used to control some types of industrial waste. One example is the use of hair to absorb heavy metals from wastewater. Hair has been shown to be an effective adsorbent for metals such as lead, copper, and zinc.



2. When hair is used to absorb heavy metals from wastewater, it works by attracting the metals to the surface of the hair fibers. This process is known as adsorption, and it can effectively remove heavy metals from the wastewater. Once the hair has absorbed the heavy metals, it can be removed and disposed of safely.
3. In addition to heavy metal adsorption, hair can also be used in composting and as a soil amendment. When hair is added to compost, it can help to increase the nutrient content of the soil, making it more fertile and able to support plant growth. The composting process can also help to break down organic matter in the waste, making it easier to manage and dispose of.

However, it's important to note that hair may not be suitable for all types of industrial waste. The effectiveness of hair as an adsorbent depends on the type and concentration of the pollutants present in the waste. Additionally, other factors such as the pH and temperature of the wastewater can also affect the effectiveness of hair as an adsorbent. So, it's essential to conduct a thorough analysis of the waste and consider the appropriate treatment options to ensure the best outcomes.

Motor Oil:

Environmental Impact:

1. When motor oil drips from vehicles, it can contaminate soil and vegetation. The oil can prevent the soil from absorbing water and can kill vegetation, which can lead

to erosion and decreased biodiversity. In addition, oil can seep into groundwater and surface water, contaminating drinking water supplies and harming aquatic life.

Economic Impact:

1. The economic impact of motor oil drips is significant. It can lead to expensive clean-up efforts, damage to infrastructure, and decreased property values. In addition, contaminated soil and water can lead to health risks and increased medical expenses.
2. 180 million gallons per year goes to oceans which is 16 times the oil spill at Exxon Valdez.

CONCLUSION

There are several possible uses of hair to reduce water pollution. Here are a few examples:

1. Absorbent material: Hair can be used as an absorbent material to remove pollutants from water. For example, hair clippings can be used to create mats or booms to absorb oil spills.
2. Filtration: Hair can be used as a filter medium to remove pollutants from water. Hair fibers can trap small particles and pollutants, making it a useful material for filtration systems.
3. Nutrient removal: Hair can also be used to remove excess nutrients from water, such as nitrogen and phosphorus, which can cause harmful algal blooms. Hair has a high nitrogen content and can be used in constructed wetlands to remove excess nutrients from water.
4. Water quality monitoring: Hair can be used as a bioindicator to monitor water quality. The presence of certain pollutants in water can affect the health of fish and other aquatic organisms, which can be reflected in the composition of their hair or scales.

Analyzing the hair or scales of fish can provide insight into the level of pollution in the water.

5. These are just a few possible uses of hair to reduce water pollution. As hair is a renewable and abundant resource, it has the potential to be an effective and sustainable solution for water pollution.

REFERENCES:

"Hair as a Low-Cost Adsorbent for Heavy Metals Removal from Wastewater: A Review," International Journal of Environmental Research and Public Health, 2018, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6056611/

"The Use of Human Hair for the Removal of Heavy Metals from Wastewater," Journal of Environmental Science and Health, Part A, 2008, https://www.tandfonline.com/doi/abs/10.1080/10934520802211631

"Sustainable use of human hair waste as a soil amendment and environmental adsorbent," Journal of Cleaner Production, 2019, https://www.sciencedirect.com/science/article/pii/S0959652619313302

"Oil Sorption by Human Hair and Wool Fibers," Journal of Environmental Engineering, 2002, https://ascelibrary.org/doi/10.1061/%28ASCE%290733-9372%282002%29128%3A11%281012%29

"Preparation and characterization of hair mats for oil spill cleanup," Journal of Hazardous Materials, 2010, https://www.sciencedirect.com/science/article/pii/S0304389410014642